

Evaluating the Accuracy of Seasonal Time Series Models for Brazilian Fuel Sales Demand

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Abstract—Accurate prediction of fuel demand is essential for energy and logistics planning in Brazil. This study evaluates the performance of SARIMA (Seasonal Auto-Regressive Integrated Moving Average) models in forecasting the monthly consumption of six types of fuels across all 27 Federative Units (UFs) of the country, totaling 162 distinct time series for each UF-Product pair. Based on the analysis of performance metrics in in-sample and out-of-sample horizons, the strengths and limitations of the model were identified. The results reveal a highly heterogeneous behavior: SARIMA achieved high accuracy in high-volume series with stable behavior, such as Gasoline C and LPG, but presented unsatisfactory performance in lower-volume or higher-volatility series, such as aviation fuels and ethanol in regional markets, where it did not outperform naïve forecasts ($TheilU \geq 1$), exhibiting negative R^2 and a systematic underestimation bias. It is concluded that, while SARIMA is effective in predictable and stable contexts, its application to irregular series presents significant limitations. To support future analyses and energy policies, an interactive Shiny application was developed to dynamically explore the results by UF and product.

Index Terms—Time series; fuel consumption; SARIMA; demand forecasting; R/Shiny

I. INTRODUCTION

The Brazilian energy matrix is marked by a relevant dependence on fossil fuels, especially petroleum derivatives and natural gas, such as gasoline, diesel oil, and liquefied petroleum gas (LPG). These energy sources are fundamental for the transport, industrial and domestic sectors, powering light, heavy, and agricultural vehicles, in addition to providing energy for residential and commercial use. Despite advances in matrix diversification driven by renewable sources such as hydroelectricity and biofuels, especially ethanol, petroleum derivatives continue to play a central role in the economy and the daily life of the population. Ensuring the efficient supply of these products is therefore strategic for the country's socioeconomic development.

In this context, understanding and anticipating fuel consumption patterns is essential for Brazil's energy and logistics

planning. Reliable projections allow for the optimization of distribution and storage, cost reduction, and the avoidance of both shortages and supply surpluses, which can generate "energy overload" a situation where generation and storage capacity exceed effective demand, implying resource waste and loss of operational efficiency. Furthermore, precise analyses of fossil fuel demand help formulate energy transition strategies, allowing the national system to prepare for the gradual introduction of clean and renewable sources in a balanced and sustainable manner.

Fuel consumption forecasting is an intricate challenge. Brazil has continental dimensions and presents great regional heterogeneity: the 27 Federative Units (UFs) differ in population density, degree of industrialization, income, and economic profile, factors that directly influence consumption volumes and seasonality. Each combination of UF and product type presents its own behavior, requiring individualized modeling approaches. Additionally, climatic, cultural, and economic factors such as vacations, agricultural harvests, and long holidays add significant seasonal fluctuations over time.

Among the most consolidated forecasting techniques for this type of data are the ARIMA (Auto-Regressive Integrated Moving Average) (Hyndman) models and their seasonal variant, the SARIMA (Seasonal ARIMA) (Hyndman). These models have proven effective in capturing both long-term trends and seasonal oscillations, being widely employed in energy demand studies. SARIMA, in particular, allows for the adjustment of monthly, quarterly or yearly periodic components, making it suitable for historical fuel consumption series where seasonality is marked.

The general objective of this work is to evaluate the performance of the SARIMA model in forecasting monthly fuel consumption in Brazil, considering six products: Hydrous Ethanol, Gasoline C (a mixture of gasoline with about 27% anhydrous ethanol), Aviation Gasoline, LPG, Diesel Oil, and Aviation Kerosene, across all 27 Federative Units. In total, 162

distinct time series ($UF \times Product$) are analyzed.

Specifically, three training and validation experiments are conducted: (I) Model trained from Jan/2010 to Dec/2023 and tested from Jan/2024 to Sep/2025 (out-of-sample 1); (II) Model trained from Jan/2010 to Sep/2024 and tested from Oct/2024 to Sep/2025 (out-of-sample 2); (III) Final model trained from Jan/2010 to Sep/2025, tested from Oct/2024 to Sep/2025 (in-sample), used in the Shiny application.

The analysis was entirely developed in R, with time series processing and modeling, and evaluation by rigorous statistical metrics (MAE, MAPE, RMSE, R^2 , and Theil's U index). Finally, an interactive Shiny application was created for dynamic visualization and comparison of results by UF and product, offering an exploratory tool aimed at improving national energy planning.

II. LITERATURE REVIEW

Demand forecasting in complex sectors, such as the fuel industry, requires models capable of capturing temporal dependencies, trends, and cyclical behavior. Among these models, ARIMA (Auto-Regressive Integrated Moving Average) stands out as one of the main statistical approaches for univariate time series. Its formulation combines three essential components: (i) Auto-Regressive (AR), which models the influence of past values of the series on the current value; (ii) Integrated (I), which applies successive differencing to remove trends and make the series stationary; and (iii) Moving Average (MA), which represents the dependence between current values and forecast errors from previous periods. This structure allows ARIMA to handle autocorrelation and non-stationarity, features commonly observed in consumption series.

However, when the series exhibits recurring patterns over time, as is the case with fuel sales throughout the year, the traditional ARIMA model becomes limited. To capture this dynamic, the SARIMA (Seasonal ARIMA) model is employed, adding specific components to represent seasonality and enabling the simultaneous modeling of short-term dependence, long-term trends, and recurring annual patterns.

This combined modeling makes SARIMA especially suitable for economic and energy series, where seasonal oscillations are linked to climatic, cultural, or productive phenomena. Furthermore, recent studies confirm its effectiveness in energy forecasting contexts, demonstrating competitive results against more complex machine learning methods when there is temporal regularity and well-structured data. [4][5].

Predictive accuracy assessment is an essential part of applying time series models. To measure the performance of models like SARIMA, statistical metrics capturing errors at different scales are widely used: MAE (Mean Absolute Error) and RMSE (Root Mean Squared Error) quantify the average magnitude of absolute and squared error, respectively; MAPE (Mean Absolute Percentage Error) expresses the error in percentage terms, facilitating comparisons between series of different magnitudes; R^2 (coefficient of determination) measures the degree of explanation of observed variation; and Theil's U index compares the model's performance with

naïve forecasts where the prediction for the next period is simply the last observed value with values $U < 1$ indicating more effective models [6].

Besides metrics, the temporal validation strategy is determinant for analytical rigor. In particular, out-of-sample validation—where the model is trained on a historical subset and tested on a subsequent period not used in fitting—is considered the most reliable form of assessing a model's real predictive power. This approach allows observing how the model generalizes to future data, reproducing the real forecasting context. On the other hand, in-sample validation measures the model's ability to fit the data that trained it, serving as a reference for potential fit, but not necessarily for generalization [7].

The combination of these two approaches provides a comprehensive view of model performance: while the out-of-sample test evaluates robustness and forecasting capacity, the in-sample test allows identifying if the model is overfitted or underfitted. This distinction is particularly important in models applied to energy forecasting, where stability and reliability of predictions have direct implications on logistics planning and public policy formulation.

In practice, the application of SARIMA models has benefited from the advancement of computational tools and statistical languages like R, which offers specialized libraries (e.g., *forecast*) to estimate and validate time series models. Furthermore, the use of interactive applications developed in Shiny expands the reach and transparency of analyses, allowing dynamic visualization of results by series, product, or region, and facilitating communication among analysts, managers, and energy policy makers.

In summary, the literature points to SARIMA as a robust, interpretable, and widely validated model for seasonal forecasts, especially when combined with good pre-processing and temporal validation practices. Its application in fuel consumption contexts offers not only quantitative forecasts but also a solid basis for energy planning and the transition to a more sustainable and efficient matrix.

III. METHODOLOGY

A. Data and Pre-Processing

The study used historical monthly fuel sales data in Brazil between January 2010 and September 2025, obtained from official records of the National Agency of Petroleum, Natural Gas and Biofuels (ANP). The series were disaggregated by Federative Unit (UF) and product type, totaling 162 distinct time series ($27 UFs \times 6$ products: Hydrous Ethanol, Gasoline C, Aviation Gasoline, Liquefied Petroleum Gas - LPG, Diesel Oil, and Aviation Kerosene). Each observation represents the monthly volume sold, in cubic meters (m^3).

During pre-processing, the following steps were performed:

1) Standardization and data structuring:

Standardization of column names, conversion of UF and Product to factors, replacement of commas with dots in numeric fields, and creation of the date variable

(Date) using the `make_date()` function from the `lubridate` package.

- 2) Filtering of products with low statistical relevance: Fuel Oil and Illuminating Kerosene were removed, as they were responsible for more than 95% of all zero records in the original set. These categories present residual volume and predominantly null behavior, which makes modeling unfeasible and distorts aggregated metrics.
- 3) Observed zeros (not missing): In the original set, about 10% of observations presented values exactly equal to zero. After excluding the two aforementioned products, this proportion reduced substantially, leaving only real zeros, compatible with periods of low or no consumption in some UFs or products. These values were kept in full, as they are considered valid and official data, not missing data. Thus, no additional imputation or removal was applied.

The result was a consistent, homogeneous, and chronologically continuous dataset, suitable for temporal modeling while preserving the integrity of official ANP records.

B. SARIMA Modeling

Each UF $\times$ Product combination was modeled individually using a univariate SARIMA model in R, with a pipeline that automates the preparation, fitting, and storage steps. The procedure consists of:

- 1) Definition of time windows:
 - The last available month (`last_month`) defines the database limit.
 - The test window comprises the final n months (`test_start` to `test_end`), while the training window (`train_start` to `train_end`) includes the rest of the history.
 - For out-of-sample experiments, training stops one month before the start of the test window ; for the final model (used in Shiny), training includes the entire period up to September 2025.
- 2) Continuous calendar guarantee:
 - The `completa_meses()` function orders the series, creates the complete monthly sequence, and preserves original values, ensuring regular frequency (12 observations per year) and no gaps.
- 3) Automatic model fitting:
 - The `fit_one()` function prepares the series for modeling, restricts training to series with at least 24 months of observations, and transforms data into monthly time series objects (ts).
 - The model is fitted with `auto.arima()` (forecast package), with `seasonal = TRUE`, `stepwise = FALSE`, and `approximation = FALSE`, exhaustively searching for the most suitable temporal and seasonal dependence structure for each series.
 - The fits are stored in a named list, where each model is indexed by the key "UF-Product".
- 4) Packaging and reproducibility:

- The results (fitted models and metadata, such as training period, frequency, and series keys) are consolidated into a list(`modelos`, `meta`) object and saved in .rds format for later use in the Shiny application.
- This artifact allows complete reproducibility of the experiment and reuse of models in interactive environments or automated routines.

In summary, each series was fitted with a SARIMA model that captures both the long-term trend and annual seasonality, checking residual adequacy before storing the final model.

C. Validation and Evaluation Windows

Predictive evaluation followed a strict temporal strategy, dividing history into non-overlapping training and testing windows. Three complementary experiments were conducted:

- 1) Out-of-sample 1: Training from Jan/2010 to Dec/2023 and testing from Jan/2024 to Sep/2025;
- 2) Out-of-sample 2: Training from Jan/2010 to Sep/2024 and testing from Oct/2024 to Sep/2025;
- 3) In-sample (final model): Training from Jan/2010 to Sep/2025 and internal testing from Oct/2024 to Sep/2025, used in the Shiny application.

For model assessment, we focused on the two experiments with 12-month forecasting horizons, as they are directly comparable. Longer horizons introduce greater variance and typically inflate predictive errors, making evaluations less stable. Concentrating on these two 12-month windows allowed us to study model behavior under equivalent temporal conditions and ensure a more reliable comparison of generalization performance.

D. Computational Environment

All steps were implemented in R language, using an integrated set of libraries:

- Dplyr and tidyr - For manipulation, filtering, and organization of tabular data;
- Lubridate - For date manipulation and monthly periodicity control;
- Forecast - For fitting and diagnosing SARIMA models;
- Ggplot2: For visualization of series and forecasts;
- Shiny: For development of the interactive results exploration application.

The combined use of these tools ensured transparency, reproducibility, and scalability. The code was modularized to enable automatic re-execution as new monthly data becomes available, ensuring continuous update of the predictive system.

IV. MODEL TRAINING

Model training followed the evaluation design described in the previous section, in which each UF-Product pair was modeled as an independent univariate time series. All implementations were carried out in R (version 4.x), primarily using the forecast package (Hyndman et al.) for fitting SARIMA models and generating predictions.

For each of the temporal experiments two out-of-sample windows and one in sample window the `auto.arima()` function

was applied individually to every training segment, selecting the most appropriate SARIMA specification according to its built-in statistical criteria. Once the optimal configuration was identified, the model was fitted to each training window and used to forecast the full horizon corresponding to that experiment. While the second out-of-sample window and the in-sample experiment involved a 12-month forecast horizon, the first out-of-sample experiment required a longer forecasting period, and the procedure was adapted accordingly to ensure consistency across all 162 UF–Product series.

Across all experiments, the full set of accuracy metrics (MAE, RMSE, MAPE, R^2 , and Theil’s U) was computed for each of the 162 series, producing three separate evaluation datasets. These metric tables allow systematic comparison of forecasting performance across different products, regions, and temporal windows. After the evaluation phase, a single final SARIMA model was trained for each UF–Product series using the complete historical data from January 2010 to September 2025, ensuring that the most recent consumption information was incorporated into the final forecasting structure.

The resulting outputs comprise (i) a final SARIMA model fitted on the full dataset for use in the Shiny application, and (ii) three performance datasets, each containing 162 rows—one for each UF–Product combination—and all accuracy metrics used in the comparative analysis. Although SARIMA models are relatively parsimonious, fitting and evaluating hundreds of independent models across multiple temporal windows is computationally demanding; however, the optimized routines in the forecast package enabled the full training and evaluation workflow to be executed efficiently within a practical time frame.

V. MODEL PERFORMANCE AND METRICS

The metrics analyzed in this section refer to Out-of-Sample Experiment 2, which constitutes the most relevant evaluation window for this project. Unlike Experiment 1 where the forecasting horizon extends far beyond twelve months, causing inflated errors due to long-term uncertainty and fewer observed values for comparison Experiment 2 offers a balanced and comparable 12-month horizon. Therefore, its metrics provide the fairest and most methodologically consistent assessment of predictive accuracy.

Table I summarizes the mean and median RMSE, MAPE, R^2 , and Theil’s U for each fuel, aggregated across the 27 UFs. The median values reflect the typical behavior of each product, while the means capture the influence of extreme cases in specific states.

The results reveal a clear separation between high-volume, regular-pattern fuels and more volatile or low-demand products. Gasoline C, LPG, and Diesel exhibit low median MAPE and positive median R^2 , along with Theil’s U below 1—indicating consistent gains over the naïve baseline. Conversely, Hydrous Ethanol and the aviation fuels show substantially higher errors and predominantly negative R^2 , highlighting the difficulty of modeling series with sparse volume, irregular seasonality, or structural breaks.

These indicators provide a concise overview of where SARIMA performs reliably and where univariate modeling becomes insufficient, guiding both interpretation and future methodological extensions.

Product	Stat	RMSE	MAPE (%)	R^2	Theil’s U
Ethanol	Mean	4505.95	15.27	-2.75	1.68
	Median	3098.71	12.86	-1.64	1.59
Gasoline C	Mean	6396.40	3.58	0.28	0.62
	Median	3229.38	3.14	0.51	0.52
LPG	Mean	1586.95	3.20	0.13	0.70
	Median	912.75	2.98	0.32	0.62
Aviation Kerosene	Mean	3017.16	19.76	-2.79	2.05
	Median	1330.41	16.65	-0.96	1.63
Aviation Gasoline	Mean	33.44	60.27	-0.23	0.81
	Median	29.52	27.66	-0.08	0.75
Diesel	Mean	12185.66	6.70	0.33	0.87
	Median	7375.53	4.84	0.53	0.68

TABLE I
SUMMARY OF FORECASTING PERFORMANCE METRICS BY PRODUCT
(OUT-OF-SAMPLE EXPERIMENT 2).

VI. INTERACTIVE APPLICATION (SHINY)

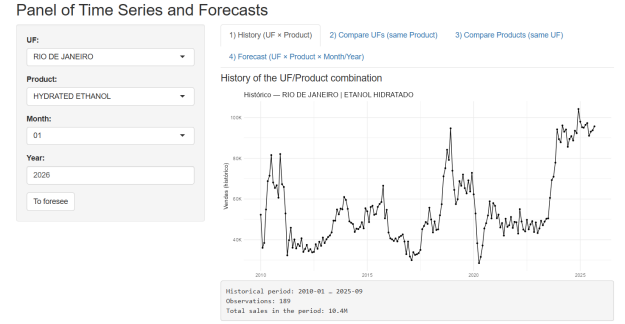


Fig. 1. User interface of the Shiny application.

As part of the study, an interactive web application was developed using the R/Shiny framework to make results available in a didactic and exploratory manner. The application serves as a complement to the static analysis presented in the article, allowing users to interact with data and forecasting outputs dynamically.

The interface is organized into a left-side control panel and a main visualization area. In the control panel, users can select the Federative Unit (UF), Product, and optionally define a future Month–Year combination to generate forward-looking projections. The main area contains four tabs, each providing a different analytical perspective through dedicated graphical displays.

- 1) **Historical Series:** This tab presents the historical time series for the selected UF–Product pair across the full training horizon (2010–2025). Data are plotted as a continuous line, allowing users to visually inspect long-term trends, seasonal patterns, and structural variations specific to the chosen series.

- 2) Comparison Across UFs: This visualization enables comparison of historical consumption patterns for a single product across multiple UFs. A set of checkboxes allows users to select which UFs to display on the same plot, facilitating cross-regional comparisons and highlighting heterogeneity in consumption dynamics.
- 3) Comparison Across Products: In this view, users can examine the behavior of different products within a single UF. As in the previous tab, checkboxes allow selecting which products to plot, enabling direct comparison of their temporal profiles and relative magnitudes over the historical period.
- 4) Future Projection: This tab provides forward-looking forecasts based on the final SARIMA models trained up to September 2025. Using the Month–Year input fields in the control panel, users can generate projections beyond the last observed date. The output graph displays forecasted values together with their 80% and 95% confidence intervals, represented through shaded areas that quantify predictive uncertainty. This visualization supports scenario exploration and illustrates how the model behaves when extended into future periods.

These four components constitute the complete functionality of the application, providing an interactive environment for analyzing temporal patterns, comparing series across regions and products, and producing future projections based on the final fitted models.

As part of the study, an interactive web application was developed using the R/Shiny framework to make results available in a didactic and exploratory manner. The application serves as a complement to the static analysis presented in the article, allowing users to interact with data and forecasting models dynamically. The main functionalities implemented include:

VII. RESULTS

The evaluation presented in this section refers specifically to Out-of-Sample Experiment 2, in which models were trained up to September 2024 and tested from October 2024 to September 2025. This evaluation window provides the fairest assessment of predictive performance, as it balances two aspects: (i) a sufficiently long and recent training history to calibrate seasonal and trend dynamics, and (ii) a 12-month forecasting horizon comparable across all UF–Product pairs. This makes the resulting metrics more homogeneous and methodologically aligned with the intended annual-cycle analysis.

Overall, SARIMA’s performance was heterogeneous across the 162 series. The median MAPE was approximately 6.4%, indicating high accuracy for at least half of the cases. However, the mean MAPE reached 20.5%, inflated by outlier series with extremely low consumption volumes most notably Aviation Gasoline, which concentrates near-zero values in several states. When these atypical series are excluded, the mean MAPE decreases to 8.6%, reflecting a more realistic depiction of model performance for regular-demand fuels.

For high-volume products exhibiting stable seasonal patterns, results were consistently strong. Gasoline C and LPG

achieved low MAPE values across nearly all UFs, averaging close to 3.5%. Diesel Oil also performed well, with a average MAPE of 6.7%, demonstrating that SARIMA effectively captured both trend and seasonal structure in these regular series. In contrast, products with irregular or highly volatile behavior especially Hydrous Ethanol and Aviation Gasoline displayed substantially higher errors, with MAPE frequently surpassing 15%, and in extreme cases exceeding 80% or even 200%.

The comparison with the naïve model (Theil’s U Index) further highlighted this division:

- 63% of series (102 cases) exhibited ($U < 1$), meaning SARIMA outperformed the naïve forecast. These gains were most prominent in Gasoline, Diesel, and LPG.
- 37% of series (60 cases) showed ($U \geq 1$), indicating no improvement over the naïve baseline. These were almost exclusively low-volume or highly volatile series, where the model could not extract a stable predictive pattern.

The out-of-sample R^2 metric reinforced these findings. 51% of models (83 series) achieved positive R^2 , indicating explanatory power relative to the mean of the test period, while the remaining (49%) produced $R^2 \leq 0$, a clear sign of failure to generalize in highly erratic series. Only Gasoline, LPG, and Diesel maintained positive average R^2 across states.

Taken together, the results indicate that SARIMA performs strongly for fuels with high volume, stable seasonality, and regular monthly behavior, while struggling in series dominated by noise, sparse consumption, or abrupt fluctuations. These findings underscore the importance of product-level heterogeneity in forecasting fuel demand and highlight which segments are most suitable for traditional time-series modeling within policy, planning, or operational contexts.

VIII. DISCUSSION

The results confirm that SARIMA models are highly effective for time series with stable consumption patterns and well defined seasonality, such as Gasoline and LPG, which aligns with findings widely reported in the literature. In these cases, the model delivers robust forecasts and achieves substantial improvements over naïve benchmarks, demonstrating its suitability for high-volume fuel demand modeling.

A central limitation and one of the key contributions of this study is SARIMA’s inability to perform well in low predictability environments. Series characterized by low volume, high volatility, or structural breaks showed markedly poor results, with nearly half of the models failing to outperform the naïve forecast. This behavior is inherent to the SARIMA framework: it extrapolates past linear and seasonal dynamics and is therefore unable to capture exogenous shocks, abrupt behavioral shifts, or patterns driven by non seasonal external influences.

The most evident methodological limitation of the study is its univariate nature. By construction, SARIMA does not incorporate exogenous determinants such as fuel prices, economic activity (e.g., GDP indicators), or contextual events like post-pandemic recovery. These variables could plausibly reduce systematic underestimation in volatile series and enhance

the model's responsiveness to real world drivers. Extending the analysis to SARIMAX or other multivariate approaches represents a clear and promising direction for future research.

Despite these constraints, the study's main strength lies in its granular scope 162 independent UF-Product series and in its transparent reporting of model shortcomings. The systematic use of metrics such as Theil's U and out of sample R^2 provides a detailed map of where SARIMA is a reliable forecasting tool and where it should not be employed in isolation for energy planning and decision-making. This contributes both methodological rigor and practical guidance for the modeling of fuel demand in heterogeneous regional contexts.

IX. CONCLUSION

This study rigorously evaluated the use of SARIMA models to forecast monthly fuel consumption by UF and product in Brazil. The results indicate that SARIMA is a viable and accurate approach for roughly half of the cases particularly for high-volume and regular pattern fuels such as Gasoline, Diesel, and LPG, where the model achieved low percentage errors (median of 9.6%) and consistently outperformed naive baselines. For these segments, SARIMA proves useful for logistical planning and short-term decision-making.

Conversely, the study also showed that SARIMA is inadequate for low-volume, highly volatile, or structurally unstable series, where performance often fell below that of a naive forecast. The contribution of this work is twofold: (i) it provides a detailed and unprecedentedly granular assessment of SARIMA accuracy across 162 UF-Product series in Brazil, and (ii) it clearly delineates the limits of the univariate approach, identifying where the model systematically fails. The Shiny application developed as part of this project further enhances practical value by translating static results into an interactive environment for exploration and scenario analysis.

Future research should consider the inclusion of exogenous variables through SARIMAX models or the adoption of hybrid frameworks that combine classical time-series structures with machine-learning techniques such as XGBoost or LSTM. These approaches may better capture nonlinearities and external drivers, potentially reducing errors in the most challenging series identified in this study.

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